

Suite Hard • Simulation Module 16

10 questions • 16 minutes • official SAT Math Hard

Exclude-active: these items are not on current Bluebook full-length tests.

All items are Hard. This is not an adaptive Module 1. It is harder than test day.

Calculator / Desmos is allowed on every item, same as Bluebook.

Question 1 is the first page after this cover. Write answers on the last page.

Read the prompt carefully. Write the job on scratch before you copy numbers.

Domain mix in this module:

Algebra: 2

Advanced Math: 3

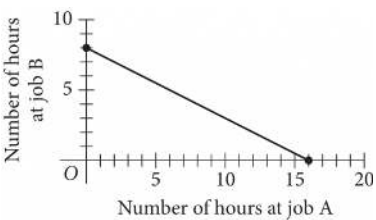
PSDA: 2

Geometry: 3

Do not open the next module until this one is timed and scored.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	Hard

Question



To earn money for college, Avery works two part-time jobs: A and B. She earns \$10 per hour working at job A and \$20 per hour working at job B. In one week, Avery earned a total of s dollars for working at the two part-time jobs. The graph above represents all possible combinations of numbers of hours Avery could have worked at the two jobs to earn s dollars. What is the value of s ?

Answer

- A. 128
- B. 160
- C. 200
- D. 320

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

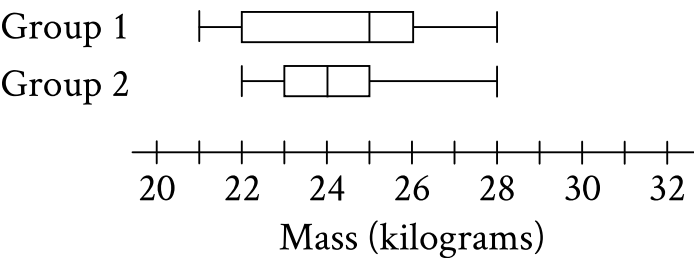
Let the function p be defined as $p(x) = \frac{(x - c)^2 + 160}{2c}$, where c is a constant. If $p(c) = 10$, what is the value of $p(12)$?

Answer

- A. 10.00
- B. 10.25
- C. 10.75
- D. 11.00

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	Hard

Question



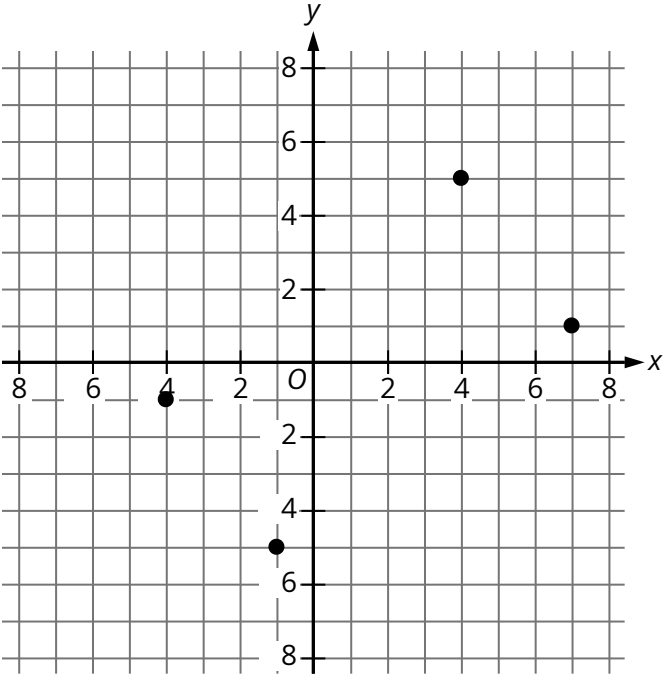
The box plots summarize the masses, in kilograms, of two groups of gazelles. Based on the box plots, which of the following statements must be true?

Answer

- A. The mean mass of group 1 is greater than the mean mass of group 2.
- B. The mean mass of group 1 is less than the mean mass of group 2.
- C. The median mass of group 1 is greater than the median mass of group 2.
- D. The median mass of group 1 is less than the median mass of group 2.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Hard

Question



A rectangle is formed by the four points shown. What is the area, in square units, of the rectangle?

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	Hard

Question

$$y = 2x + 1$$
$$y = ax - 8$$

In the system of equations above, a is a constant. If the system of equations has no solution, what is the value of a ?

Answer

- A. $-\frac{1}{2}$
- B. 0
- C. 1
- D. 2

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	Hard

Question

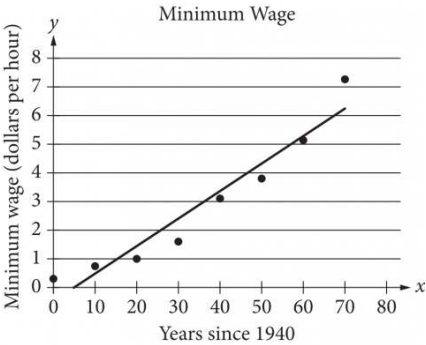
Which of the following expressions is equivalent to $\frac{x^2-2x-5}{x-3}$?

Answer

- A. $x - 5 - \frac{20}{x-3}$
- B. $x - 5 - \frac{10}{x-3}$
- C. $x + 1 - \frac{8}{x-3}$
- D. $x + 1 - \frac{2}{x-3}$

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Two-variable data: Models and scatterplots	Hard

Question



The scatterplot above shows the federal-mandated minimum wage every 10 years between 1940 and 2010. A line of best fit is shown, and its equation is $y = 0.096x - 0.488$. What does the line of best fit predict about the increase in the minimum wage over the 70-year period?

Answer

- A. Each year between 1940 and 2010, the average increase in minimum wage was 0.096 dollars.
- B. Each year between 1940 and 2010, the average increase in minimum wage was 0.49 dollars.
- C. Every 10 years between 1940 and 2010, the average increase in minimum wage was 0.096 dollars.
- D. Every 10 years between 1940 and 2010, the average increase in minimum wage was 0.488 dollars.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	Hard

Question

A circle in the xy-plane has equation $(x + 3)^2 + (y - 1)^2 = 25$. Which of the following points does NOT lie in the interior of the circle?

Answer

- A. $(-7, 3)$
- B. $(-3, 1)$
- C. $(0, 0)$
- D. $(3, 2)$

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Hard

Question

$-2x^2 + 20x + c = 0$

In the given equation, c is a constant. The equation has exactly one solution. What is the value of c ?

Answer

- A. **−68**
- B. −50
- C. **−32**
- D. 0

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	Hard

Question

What is the perimeter of an equilateral triangle with a height of $39\sqrt{3}$?

Answer

- A. 117
- B. $117\sqrt{3}$
- C. 234
- D. $1,521\sqrt{3}$

Module 16 • answer sheet

Write one answer per line. SPR: integer, decimal, or fraction.

1. _____
2. _____
3. _____
4. _____
5. _____
6. _____
7. _____
8. _____
9. _____
10. _____